

25 prompts for data scientists and BI professionals

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PROMPTS

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Data & Analytics

Works with: **ChatGPT** • **Claude** • **Gemini** • **Llama**

About This Playbook

25 expert prompts covering SQL, exploratory analysis, A/B testing, dashboards, reporting, and data storytelling — for analysts who want to work smarter.

What You'll Get

• 25 expert prompts • Professional frameworks • Placeholder system • Expert tips per prompt

Who It's For

• Entrepreneurs • Professionals • Teams • Freelancers

How to Use This Playbook

- 1 Copy the prompt — Select the full prompt text from the section that matches your goal.
- 2 Fill in the brackets — Replace all [PLACEHOLDER] text with your specific context, brand, or product.
- 3 Paste & generate — Paste into ChatGPT, Claude, Gemini, or your preferred AI tool.
- 4 Iterate — Use the output as a starting point. Refine with follow-up prompts for best results.

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Section 1

Section 1 (5 prompts)

Prompt #01

SQL Query Generator

Write optimized SQL queries for [DATABASE] to answer [BUSINESS QUESTION]

You are a senior data engineer. Write optimized SQL queries for [DATABASE] to answer [BUSINESS QUESTION]. Deliver: 1. The query (readable, commented, optimized) 2. Explanation of logic 3. Performance notes (indexes, query plan) 4. 3 variations for edge cases

Placeholders: [DATABASE] e.g. PostgreSQL, BigQuery [BUSINESS QUESTION] What you're analyzing

■ **Tip:** Add comments to every non-obvious SQL line. Your future self will thank you.

ChatGPT ✓

Claude ✓

Gemini ✓

Prompt #02

Window Functions Masterclass

Write 5 window function queries for [DATASET] covering ranking, running

You are a SQL expert. Write 5 window function queries for [DATASET] covering ranking, running totals, and lag/lead analysis. Deliver: For each query: Business question + SQL + explanation + when to use vs alternatives

Placeholders: [DATASET] Your data source

■ **Tip:** Window functions replace 80% of self-joins. Learn them and your queries get 5x cleaner.

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Gemini ✓

Prompt #03

EDA Framework — Exploratory Data Analysis

Run a complete EDA on [DATASET] and report findings.

You are a data scientist. Run a complete EDA on [DATASET] and report findings. Deliver: 1. Data shape and types 2. Missing value analysis and imputation strategy 3. Distribution analysis (numerical) 4. Frequency analysis (categorical) 5. Correlation matrix and key relationships 6. Anomaly/outlier detection 7. Hypotheses for further investigation

Placeholders: [DATASET] Your dataset description

■ **Tip:** EDA is not cleaning — it's understanding. Ask why before deciding how to handle anomalies.

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Prompt #04

Data Cleaning Checklist

Create a data cleaning plan for [DATASET] before analysis.

You are a data quality specialist. Create a data cleaning plan for [DATASET] before analysis. Deliver: 1. Duplicate removal strategy 2. Missing value handling (drop/impute/flag) 3. Outlier treatment policy 4. Data type standardization 5. String normalization (case, whitespace, special chars) 6. Date/time parsing 7. Referential integrity checks

Placeholders: [DATASET] Your data

■ **Tip:** Document every cleaning decision. Undocumented cleaning creates unreproducible analysis.

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Prompt #05

Cohort Analysis Framework

Build a cohort analysis for [PRODUCT/SERVICE] measuring [METRIC] over

You are a growth analyst. Build a cohort analysis for [PRODUCT/SERVICE] measuring [METRIC] over [TIME PERIOD]. Deliver: 1. Cohort definition (acquisition date grouping) 2. Cohort table structure 3. SQL query for cohort retention 4. Visualization recommendations 5. Interpretation: What patterns to look for

Placeholders: [PRODUCT/SERVICE] What you're analyzing [METRIC] e.g. retention, revenue [TIME PERIOD] e.g. 12 months

■ **Tip:** Cohort analysis reveals if you're getting better at retaining customers over time.

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Section 2

Section 2 (5 prompts)

Prompt #06

A/B Test Statistical Analysis

Analyze A/B test results for [EXPERIMENT] with statistical rigor.

You are a quantitative analyst. Analyze A/B test results for [EXPERIMENT] with statistical rigor. Deliver: 1. Hypothesis statement (null and alternative) 2. Test selection (t-test, chi-square, Mann-Whitney) 3. Sample size adequacy check 4. Statistical significance calculation 5. Effect size interpretation 6. Practical significance vs statistical significance 7. Decision recommendation

Placeholders: [EXPERIMENT] Your A/B test description

■ **Tip:** Statistical significance $\neq$ practical significance. A 0.1% improvement can be statistically significant but business-irrelevant.

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Prompt #07

Funnel Analysis Query

Write a funnel analysis for [CONVERSION FUNNEL] in [DATA TOOL].

You are a conversion analyst. Write a funnel analysis for [CONVERSION FUNNEL] in [DATA TOOL]. Deliver: 1. Step-by-step funnel query 2. Drop-off rate calculation per step 3. Cohort breakdown of funnel performance 4. Segmentation: Which segments have highest/lowest conversion? 5. Time-to-convert analysis

Placeholders: [CONVERSION FUNNEL] e.g. signup → activation → paid [DATA TOOL] SQL/Python/Amplitude

■ **Tip:** The biggest drop-off step is almost always your highest ROI optimization opportunity.

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Prompt #08

Executive Dashboard Design

Design a KPI dashboard for [BUSINESS UNIT/PRODUCT].

You are a BI specialist. Design a KPI dashboard for [BUSINESS UNIT/PRODUCT]. Deliver: 1. KPI selection (max 7 per view – fewer is better) 2. Hierarchy: North Star → leading indicators → diagnostic metrics 3. Visualization type for each KPI 4. Update frequency and data freshness requirements 5. Alert thresholds 6. SQL queries for each metric

Placeholders: [BUSINESS UNIT/PRODUCT] What you're measuring

■ **Tip:** Every metric on a dashboard should be actionable. If you can't act on it, remove it.

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Claude ✓

Gemini ✓

Prompt #09

Customer Segmentation Model

Build a customer segmentation model for [BUSINESS] using [DATA].

You are a data science practitioner. Build a customer segmentation model for [BUSINESS] using [DATA]. Deliver: 1. Segmentation approach (RFM, behavioral, demographic, psychographic) 2. Feature selection and engineering 3. Clustering methodology 4. Segment profiles (4-6 segments) 5. Marketing implication for each segment

Placeholders: [BUSINESS] Your business [DATA] Available data

■ **Tip:** Start with RFM segmentation. It's simple, interpretable, and immediately actionable.

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Prompt #10

Marketing Attribution Analysis

Build an attribution model for [MARKETING CHANNELS] for [BUSINESS].

You are a marketing analyst. Build an attribution model for [MARKETING CHANNELS] for [BUSINESS]. Deliver: 1. Attribution model options (first-touch, last-touch, linear, time-decay, data-driven) 2. Implementation for [CHANNELS] 3. SQL for multi-touch attribution 4. Comparison of models and what each one rewards 5. Recommendation for [BUSINESS TYPE]

Placeholders: [MARKETING CHANNELS] List your channels [BUSINESS] Your business

■ **Tip:** No attribution model is perfect. Use multiple and triangulate. Last-click is always wrong.

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Gemini ✓

Section 3

Section 3 (5 prompts)

Prompt #11

Churn Prediction Feature Engineering

Build churn prediction features for [PRODUCT].

You are a machine learning engineer. Build churn prediction features for [PRODUCT]. Deliver: 1. Target variable definition 2. Feature categories (behavioral, demographic, engagement, support) 3. Specific feature engineering for each category 4. Feature importance methodology 5. Data pipeline design

Placeholders: [PRODUCT] What you're building for

■ **Tip:** Churn prediction models are only as good as your feature engineering. Behavioral features beat demographic ones.

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Prompt #12

Revenue Metrics Dashboard

Build a revenue metrics framework for [BUSINESS MODEL].

You are a revenue analyst. Build a revenue metrics framework for [BUSINESS MODEL]. Deliver: Metrics to track and calculate: MRR/ARR, Churn (logo + revenue), NRR, CAC by channel, LTV, LTV:CAC, Payback period, Expansion MRR For each: Formula + definition + target benchmark + SQL

Placeholders: [BUSINESS MODEL] SaaS/e-commerce/marketplace

■ **Tip:** Your revenue metrics tell the story investors read. Know every number and what moves it.

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Prompt #13

Python Pandas Analysis Script

Write a complete Pandas analysis script for [DATASET] answering [QUEST

You are a Python data analyst. Write a complete Pandas analysis script for [DATASET] answering [QUESTIONS]. Deliver: ```python import pandas as pd import numpy as np # 1. Load and inspect # 2. Clean and prepare # 3. Analyze: answer each question # 4. Visualize key findings # 5. Export results ``` With comments explaining each step

Placeholders: [DATASET] Your data file [QUESTIONS] Analysis questions

■ **Tip:** Write analysis scripts as if you'll hand them to a junior analyst in 6 months. Comment everything.

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Prompt #14

Data Storytelling Report

Transform [ANALYSIS RESULTS] into an executive-friendly story.

You are a data storyteller. Transform [ANALYSIS RESULTS] into an executive-friendly story. Deliver: 1. The headline finding (single sentence, so what?) 2. Context (why this matters now) 3. What the data shows (3 key findings) 4. Root cause analysis 5. Recommendation with supporting evidence 6. Next steps + success metrics

Placeholders: [ANALYSIS RESULTS] Your data findings

■ **Tip:** Data without narrative is noise. Your job as analyst is to make the data speak.

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Prompt #15

Google Analytics 4 Audit

Audit [WEBSITE]'s GA4 setup and provide optimization recommendations.

You are a web analytics specialist. Audit [WEBSITE]'s GA4 setup and provide optimization recommendations. Deliver: 1. Event tracking audit (what's missing?) 2. Conversion tracking verification 3. Custom dimensions and metrics needed 4. Audience building for remarketing 5. Key reports to set up 6. Data quality issues to fix

Placeholders: [WEBSITE] Your website

■ **Tip:** Most GA4 implementations track pageviews but miss events. Events are where the insights are.

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Section 4

Section 4 (5 prompts)

Prompt #16

Data Warehouse Design

Design a data warehouse architecture for [BUSINESS] with [DATA SOURCES]

You are a data engineer. Design a data warehouse architecture for [BUSINESS] with [DATA SOURCES]. Deliver: 1. Source system inventory 2. Medallion architecture (Bronze/Silver/Gold layers) 3. Key dimension and fact tables 4. Data freshness requirements 5. dbt model structure 6. Governance and access control

Placeholders: [BUSINESS] Your company [DATA SOURCES] Data systems

■ **Tip:** Design data models for query performance, not source system purity.

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Prompt #17

KPI Definition Framework

Define KPIs for [DEPARTMENT/PRODUCT] at [COMPANY].

You are a metrics strategist. Define KPIs for [DEPARTMENT/PRODUCT] at [COMPANY]. Deliver: 1. OKR → KPI mapping 2. For each KPI: Definition, formula, data source, owner, frequency, target 3. Leading vs lagging indicator balance 4. KPI hierarchy (North Star → secondary → diagnostic) 5. Review cadence

Placeholders: [DEPARTMENT/PRODUCT] What you're measuring [COMPANY] Your company

■ **Tip:** Every KPI must have an owner. Shared KPIs are orphaned KPIs.

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Prompt #18

Data Quality Monitoring Framework

Build a data quality monitoring framework for [DATA PIPELINE].

You are a data reliability engineer. Build a data quality monitoring framework for [DATA PIPELINE]. Deliver: 1. Quality dimensions: Completeness, Accuracy, Consistency, Timeliness 2. Automated tests for each dimension 3. Alert thresholds and escalation 4. Data quality scorecard 5. Incident response for data issues

Placeholders: [DATA PIPELINE] Your pipeline

■ **Tip:** Bad data costs 10x more to fix downstream than at the source. Monitor quality at ingestion.

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Prompt #19

Visualization Best Practices Guide

Write visualization recommendations for [DATA TYPE] at [COMPANY].

You are a data visualization specialist. Write visualization recommendations for [DATA TYPE] at [COMPANY]. Deliver: Which chart for which data type + when to use each visualization + color theory basics + accessibility requirements + common visualization mistakes

Placeholders: [DATA TYPE] Your data [COMPANY] Your audience

■ **Tip:** The best visualization is the one that makes the insight unmissable. Not the most complex.

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Prompt #20

Predictive Analytics Framework

Design a predictive analytics program for [BUSINESS PROBLEM] at [COMPANY]

You are a data scientist. Design a predictive analytics program for [BUSINESS PROBLEM] at [COMPANY]. Deliver: 1. Problem framing (prediction target, success metric) 2. Data requirements 3. Model selection approach 4. Validation methodology 5. Deployment and monitoring plan 6. Business integration

Placeholders: [BUSINESS PROBLEM] What to predict [COMPANY] Your company

■ **Tip:** Prediction without a deployment plan is academic. Design for how the model changes a decision.

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Section 5

Section 5 (5 prompts)

Prompt #21

Time Series Analysis Framework

Analyze [TIME SERIES METRIC] for [BUSINESS] and forecast [PERIOD] ahead

You are a time series analyst. Analyze [TIME SERIES METRIC] for [BUSINESS] and forecast [PERIOD] ahead. Deliver: 1. Decomposition: Trend + seasonality + residual 2. Stationarity testing 3. Model selection: ARIMA/ETS/Prophet 4. Forecast with confidence intervals 5. What drives the metric 6. Scenario modeling

Placeholders: [TIME SERIES METRIC] Your metric [BUSINESS] Context [PERIOD] Forecast window

■ **Tip:** Always plot the raw time series before modeling. Visual inspection catches 80% of issues.

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Prompt #22

Product Analytics Framework

Build a product analytics framework for [PRODUCT].

You are a product analyst. Build a product analytics framework for [PRODUCT]. Deliver: 1. Activation metric and funnel 2. Engagement metrics (DAU/WAU/MAU, stickiness) 3. Retention cohort setup 4. Feature adoption tracking 5. User journey analysis 6. Health score design

Placeholders: [PRODUCT] Your product

■ **Tip:** Product analytics only matters if product decisions change based on the data.

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Prompt #23

Business Intelligence Reporting Cadence

Design the reporting cadence for [COMPANY]'s analytics program.

You are a BI manager. Design the reporting cadence for [COMPANY]'s analytics program. Deliver: Daily: Operational metrics (who, what format, delivery method) Weekly: Performance review (structure, attendees, decisions) Monthly: Strategic metrics (board-level, format) Quarterly: Deep dives (topics, ownership) + Self-serve vs analyst-produced guidance

Placeholders: [COMPANY] Your company

■ **Tip:** The best analytics programs have clear reporting SLAs. Know when each report goes out and who uses it.

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Prompt #24

Data Team ROI Calculator

Calculate and communicate the ROI of the data team for [COMPANY].

You are a data strategy consultant. Calculate and communicate the ROI of the data team for [COMPANY]. Deliver: 1. Value categories: Revenue impact, cost reduction, risk reduction, speed improvement 2. Attribution methodology 3. Sample ROI calculations 4. How to present to leadership 5. Data team cost vs value comparison

Placeholders: [COMPANY] Your company

■ **Tip:** Data teams that can't articulate their ROI don't get budget. Build the business case proactively.

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Prompt #25

Self-Serve Analytics Setup

Set up self-serve analytics for [COMPANY] so non-analysts can answer t

You are a data democratization specialist. Set up self-serve analytics for [COMPANY] so non-analysts can answer their own questions. Deliver: 1. Tool selection (Metabase/Tableau/Looker/PowerBI) 2. Semantic layer design 3. Training program for business users 4. Guardrails (what they can't break) 5. Escalation to analysts 6. Documentation framework

Placeholders: [COMPANY] Your company

■ **Tip:** Self-serve analytics frees analysts from repetitive report requests. Invest in it.

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Section 6

Section 6 (1 prompts)

Prompt #26

SQL Anti-Patterns and Fixes

Identify and fix common SQL anti-patterns in [DATABASE] queries.

You are a database optimization expert. Identify and fix common SQL anti-patterns in [DATABASE] queries. Deliver: 10 anti-patterns to cover: SELECT *, N+1 queries, not using indexes, correlated subqueries, OR in WHERE, not using EXPLAIN, implicit type conversion, COUNT(DISTINCT) at scale, cross joins, non-sargable predicates For each: Example bad query + fixed query + explanation

Placeholders: [DATABASE] Your database

■ **Tip:** SQL anti-patterns are the root cause of 80% of data warehouse performance problems.

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